

# Static Gesture Classification — Run Report

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## Data paths

|                       |                                                                        |
|-----------------------|------------------------------------------------------------------------|
| Training data path 1  | /home/jestin/ThesisRepo/ML/NewTestData/1_Alan/Static · 35 trials       |
| Training data path 2  | /home/jestin/ThesisRepo/ML/NewTestData/2_Alex/Static · 35 trials       |
| Training data path 3  | /home/jestin/ThesisRepo/ML/NewTestData/3_Anghad/Static · 35 trials     |
| Training data path 4  | /home/jestin/ThesisRepo/ML/NewTestData/4_Daniel/Static · 35 trials     |
| Training data path 5  | /home/jestin/ThesisRepo/ML/NewTestData/5_Georgia/Static · 35 trials    |
| Training data path 6  | /home/jestin/ThesisRepo/ML/NewTestData/6_Harry/Static · 35 trials      |
| Training data path 7  | /home/jestin/ThesisRepo/ML/NewTestData/7_Henry/Static · 35 trials      |
| Training data path 8  | /home/jestin/ThesisRepo/ML/NewTestData/8_Jestin/Static · 36 trials     |
| Training data path 9  | /home/jestin/ThesisRepo/ML/NewTestData/9_Joselyn/Static · 30 trials    |
| Training data path 10 | /home/jestin/ThesisRepo/ML/NewTestData/10_Josh/Static · 35 trials      |
| Training data path 11 | /home/jestin/ThesisRepo/ML/NewTestData/11_Mansh/Static · 36 trials     |
| Training data path 12 | /home/jestin/ThesisRepo/ML/NewTestData/12_Marcus/Static · 35 trials    |
| Training data path 13 | /home/jestin/ThesisRepo/ML/NewTestData/14_StephenV2/Static · 35 trials |
| Training data path 14 | /home/jestin/ThesisRepo/ML/NewTestData/16_Bella/Static · 35 trials     |
| Total training paths  | 14 · 487 trials total                                                  |
| External test root    | /home/jestin/ThesisRepo/ML/NewTestData/13_Nat/Static                   |
| Report output dir     | /home/jestin/ThesisRepo/ML/NewReports/StaticReports                    |

## Sensor selection

|                |                                                                                                                     |
|----------------|---------------------------------------------------------------------------------------------------------------------|
| Hands          | left, right                                                                                                         |
| Segments       | thumb, index, middle, ring, pinky, wrist                                                                            |
| Modalities     | YPR, Accel, Flex                                                                                                    |
| Sensor columns | 82 channels                                                                                                         |
| Classes        | 7: Double_Closed_Fist, Double_Nothing, Double_Okay, Double_Open_Palm, Double_Phone, Double_Pistol, Double_Spiderman |

## Sensor grid (✓ enabled, ✗ disabled, — not present)

| Hand / Segment | IMU_mid | IMU_prox | Flex_MCP | Flex_PIP |
|----------------|---------|----------|----------|----------|
| left palm      | ✗       | ✓        | —        | —        |
| left thumb     | ✗       | ✓        | ✗        | ✓        |
| left index     | ✗       | ✓        | ✗        | ✓        |
| left middle    | ✗       | ✓        | ✗        | ✓        |
| left ring      | ✗       | ✓        | ✗        | ✓        |
| left pinky     | ✗       | ✓        | ✗        | ✓        |
| left wrist     | ✗       | —        | —        | —        |
| right palm     | ✗       | ✓        | —        | —        |
| right thumb    | ✗       | ✓        | ✗        | ✓        |
| right index    | ✗       | ✓        | ✗        | ✓        |
| right middle   | ✗       | ✓        | ✗        | ✓        |
| right ring     | ✗       | ✓        | ✗        | ✓        |
| right pinky    | ✗       | ✓        | ✗        | ✓        |
| right wrist    | ✗       | —        | —        | —        |

Enabled: 22 · Disabled: 24 (channels per enabled IMU depend on modality flags above).

## Preprocessing, features & split

|                             |                                            |
|-----------------------------|--------------------------------------------|
| Resample steps              | 90                                         |
| Butterworth                 | on · cutoff 10.0 Hz · order 4 · fs 30.0 Hz |
| Normalisation               | minmax                                     |
| Feature mode                | stats+fft → feature dim 4428               |
| Test size · seed · CV folds | 0.2 · random_state=42 · 5-fold stratified  |
| Sample counts               | train (post-aug)=1945 · test=98            |

## Augmentation (training only)

|                        |                        |
|------------------------|------------------------|
| Enabled                | yes                    |
| Copies per train trial | 4                      |
| Jitter                 | off · sigma=0.01       |
| Amplitude scale        | on · range=(0.9, 1.1)  |
| Time warp              | off · range=(0.9, 1.1) |

## Classifiers — cross-validation & hold-out test

CV = 5-fold cross-validation. Test = hold-out accuracy. M-\* = macro-averaged, W-\* = support-weighted. Prec / Rec / F1 on hold-out test.

| Algorithm | CV acc | CV std | CV F1 | CV F1<br>σ | Test | M-Prec | M-Rec | M-F1 | W-Pre<br>c | W-Rec | W-F1 |
|-----------|--------|--------|-------|------------|------|--------|-------|------|------------|-------|------|
|-----------|--------|--------|-------|------------|------|--------|-------|------|------------|-------|------|

|               |        |        |        |        |        |        |        |        |        |        |        |
|---------------|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|
| Random Forest | 0.9974 | 0.0023 | 0.9975 | 0.0023 | 0.9286 | 0.9361 | 0.9239 | 0.9186 | 0.9349 | 0.9286 | 0.9211 |
|---------------|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|

Per-class metrics (hold-out test)

=== Random Forest ===

|                    | precision | recall | f1-score | support |
|--------------------|-----------|--------|----------|---------|
| Double_Closed_Fist | 0.88      | 1.00   | 0.94     | 15      |
| Double_Nothing     | 1.00      | 0.54   | 0.70     | 13      |
| Double_Okay        | 0.93      | 0.93   | 0.93     | 14      |
| Double_Open_Palm   | 1.00      | 1.00   | 1.00     | 14      |
| Double_Phone       | 0.93      | 1.00   | 0.97     | 14      |
| Double_Pistol      | 0.93      | 1.00   | 0.97     | 14      |
| Double_Spiderman   | 0.88      | 1.00   | 0.93     | 14      |
| accuracy           |           |        | 0.93     | 98      |
| macro avg          | 0.94      | 0.92   | 0.92     | 98      |
| weighted avg       | 0.93      | 0.93   | 0.92     | 98      |

External test (NewTestData)

| Algorithm     | External test acc |
|---------------|-------------------|
| Random Forest | 0.9459            |

Per-class external accuracy

| Class              | Random Forest | Support |
|--------------------|---------------|---------|
| Double_Closed_Fist | 1.000         | 6       |
| Double_Nothing     | 0.600         | 5       |
| Double_Okay        | 1.000         | 5       |
| Double_Open_Palm   | 1.000         | 5       |
| Double_Phone       | 1.000         | 5       |
| Double_Pistol      | 1.000         | 6       |
| Double_Spiderman   | 1.000         | 5       |